

1 CardLink

1.1 Overview

When using KABA offline components (electronic door mountings and digital cylinders), CardLink allows for loading authorizations onto the badge by using the terminal program AIP or ctwin at PZE terminals. The advantage of this is that in the event of changes to the authorizations, it is not necessary to load the authorizations onto the affected components on site.

When clocking at a PZE terminal, the authorizations on the badge are automatically "validated". A time stamp is updated on the badge, and the authorizations on the badge are subsequently valid for the validation period set in B-COMM. Typically, this period is set to approx. 16 hours, so that there will not be any valid authorizations in the course of the next day for lost badges deactivated in HYDRA, and authorizations also can no longer be validated or loaded at the PZE terminal.

If the PZE terminal detects that the authorizations on the badge are no longer valid, the employee will be informed at the PZE terminal that new authorizations are available and should be loaded onto the badge.

As for the following changes, the system will recognize that the authorizations for offline components have changed:

- **Change in access profile**

A change in the access profile affects all badges to which this access profile is assigned, provided this change affects an access group which includes offline components.

- **Change in access profile assignment**

If the access profile assignments for a badge change and an access configured as an offline component is affected by this.

- **Change in validity period of a badge**

If the validity period of a badge was changed or the badge is deactivated. This may also be initiated by a change of the badge number or the date of leaving in the HR master data.



If several versions with interrupted validity periods exist for a badge in HYDRA, the authorizations are only written onto the badge until the end of the current validity period. Upon expiry of this validity period, the employee is requested to reload the authorizations at the PZE terminal.

1.2 Requirements

The use of CardLink requires memory space on the badge. Legic badges require a separate segment.

A Legic badge must be structured as follows:

- The badge must have two segments: The standard PZE/ZKS access segment and the CardLink segment.
- Each segment must have its own, unambiguous search string.
- The first segment is, by definition, the standard PZE/ZKS access segment with search string + badge data (standard MPDV segment). For this segment, reading access only is required.
- The second segment is the CardLink segment. This segment must have sufficient memory space and allow for both reading and writing access.

At present, CardLink is only available for Legic badges (Legic Prime and Legic Advant).

In addition, a special write-capable reader is required at those PZE terminals where authorizations are to be written on the badge. This is a new LEGIC Advant write-capable reader (with "LGA" in the MPDV product description; example: 382-ILGAL).

1.3 Memory requirements on the legic badge

The required size of the segment is calculated according to the following formula:

$$10 \text{ bytes} + 10 \text{ bytes} \times \text{number of authorized access groups}$$

If a badge is authorized for offline components, which are categorized in 5 different access groups, the memory space requirement is:

$$10 \text{ bytes} + 10 \text{ bytes} \times 5 = 60 \text{ bytes}$$

If the badge is authorized for an access group for several, interrupted periods, another 10 bytes are added for each period.



The required memory space for authorizations has an effect on the duration needed to write authorizations: Approx. 1 second is needed for each 100 bytes. For this reason, you should attempt to summarize the offline component accesses in as few access groups as possible.

1.4 Updating authorizations at the PZE terminal

The PZE terminals load the authorizations for offline components in a cycle of 5 minutes, so that these authorizations can also be validated and/or loaded in the offline case. This cycle duration can be set in seconds in the configuration file `hytnrcfg.ini` for terminals with the terminal program AIP, and in the file `hytnrcfg.bsp` for terminals with the terminal program ctwin, respectively:

```
[ CARDLINK.LST ]  
loadtime=300
```

1.5 Writing authorizations at the PZE terminal

The authorizations of a person are written on the badge at the PZE terminal by using one of the 4 absence reason buttons.

Required program statuses:

Terminal program	Versions
drv_crypt.dll	V# 2.0.0.2
ctwin.exe	V# 7.2.7.19
ctaip.exe .\packets\pzezks72.dll	V# 2.0.3.7 V# 2.0.1.22

1.5.1 Terminal configuration at the console

The CardLink function is configured in the [terminal configuration](#) in the tab "HR functions". The writing of authorizations on the badge can be configured with the "Absence reason" **CL** and an appropriate text on an absence reason button.

1.5.2 Terminal configuration

For writing on badges, a reader-specific DLL has to be configured at the terminal. At present, the following DLL are available:

DLL	Description
drv_crypt.dll	Driver for LEGIC Advant reader (e.g. 382-ILGAL, CTB-LGALU, CT-LGALTU)

The configuration is performed at the terminal in the file **ctaip.ini** for AIP terminals, and in the file **ctwin.ini** for terminals with the ctwin terminal program, respectively.

Driver activation (x stands for the number of the serial interface):

```
[COMPORTS]
COMx=drv_crypt
```

Activating the CardLink functionality:

```
[COMPORTS-PARAM]
DRV_CRYPT-PARAM=SETTINGS=CL |
```

Customer-specific configuration (example):

```
[COMPORTS-PARAM]
DRV_CRYPT-PARAM=SETTINGS=CL | SEARCHSTRING=2C2D2E000000 | STARTADDRESS=6 |
```

Depending on the structure of the badges used, a customer-specific configuration may also be required. Primarily, this includes settings for the CardLink segment: search string, segment size and data start address.



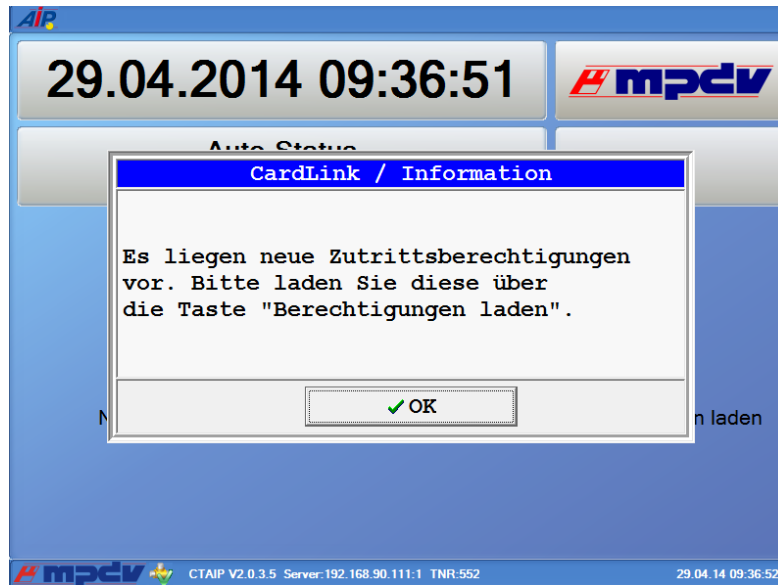
These settings have an impact on the reader behavior. These entries should be made carefully, since they have an effect on the overall behavior of the CardLink application. Adaptations should only be made after consultation with MPDV.

The parameters described below can be set within the parameter string "DRV_CRYPT-PARAM" in order to implement the customer-specific configuration. If several parameters are used, they are to be separated by "|" (Pipe).

Parameter	Description
SEARCHSTRING	Search string of the CardLink segment The default is "2C2D2E000000"
STARTADDRESS	This indicates the start address on the badge within the access segment. Usually immediately behind the search string. Example: The entry 3 corresponds to the fourth byte as start address (counting starts at 0). The authorizations are consequently written as from the fourth byte. The default value is 6.

1.5.3 Process at the terminal

New rights are available for a badge:



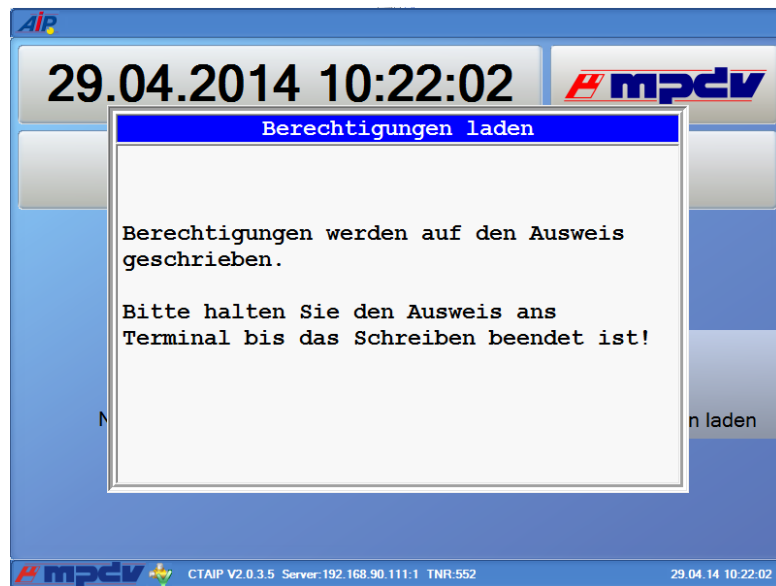
The function "Load authorizations" is selected via the relevant function key.



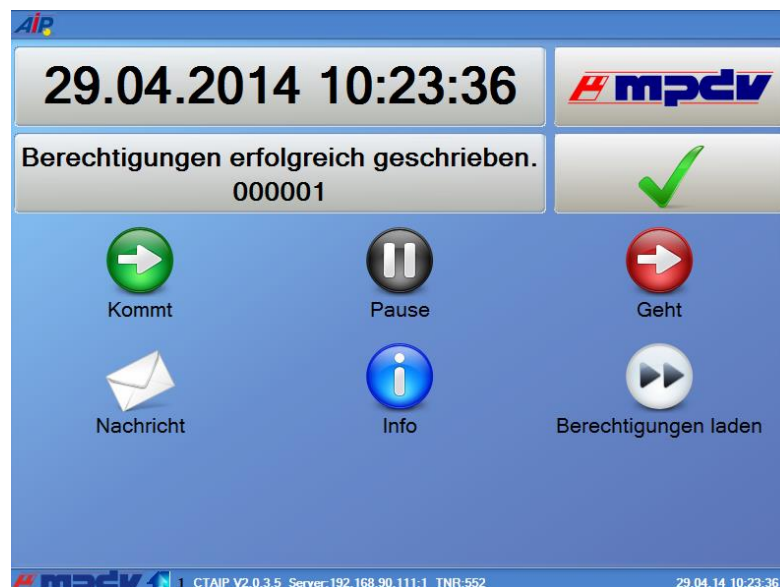
Please note:

The screenshots show the display of the AIP terminal when loading authorizations on a badge. The display at CTWIN is comparable in terms of the contents.

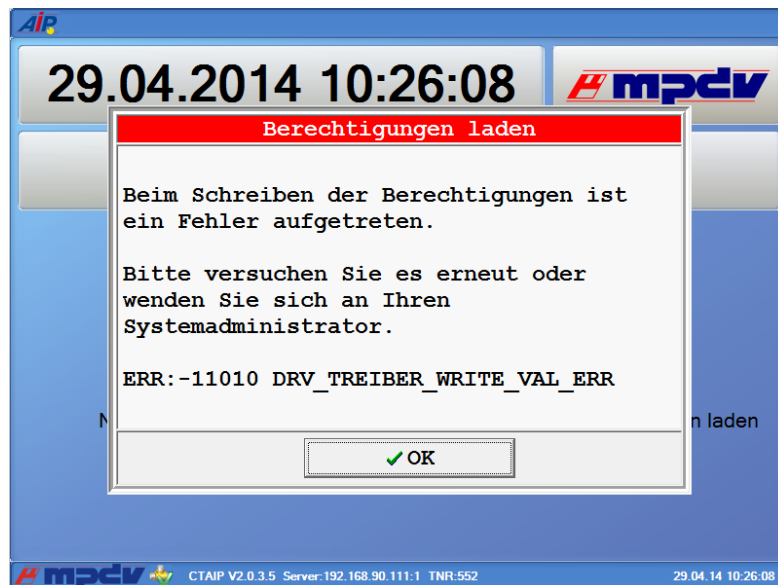
The following note is displayed when authorizations are written on a badge:



After the successful writing of authorizations, the note is closed and successful processing is indicated as follows, as when clocking:



In the case of an error, a message window opens, which can be closed by selecting "OK". The contents depend on the error occurred.



Possible errors are:

Error	Description
-11008	DRV_TREIBER_ERROR_WRITEDATA Error while writing the badge data. Possible causes: - Badge removed from reader during writing. - Badge does not have sufficient memory space. - Communication with reader and/or badge disrupted.
-11009	DRV_TREIBER_TIMEOUT_WRITEDATA Timeout while writing the badge data. Possible causes: - Communication with reader and/or badge disrupted.
-11010	DRV_TREIBER_WRITE_VAL_ERR Error during validation of badge. Possible causes: - Badge removed from reader during writing. - Communication with reader and/or badge disrupted.
-11011	DRV_TREIBER_WRITE_DEVAL_ERR Error during invalidation of badge. Possible causes: - Badge removed from reader during writing. - Communication with reader and/or badge disrupted.
-11013	DRV_TREIBER_WRITE_DATA_SEG_TO_SMALL_VALID The data to be written does not fit into the CardLink segment. Possible causes:

	- The CardLink segment is too small. - Authorizations have to be optimized.
-11014	DRV_TREIBER_WRITE_DATA_SEG_TO_SMALL_NOT_VALID The data to be written does not fit into the CardLink segment. Possible causes: - The CardLink segment is too small. Authorizations have to be optimized.

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